A COMPLEX STABLE PHASE RETRIEVAL THEOREM

Abstract. We study stable phase retrieval for spans of independent complex random variables. The starting point is the compactness strategy from the real theory: if a uniform orthogonal lower bound failed, one could extract a limit model in which two independent-coordinate sums X and Y satisfy $|X| = |Y|$ almost surely. Over the real field this immediately forces a global sign; over the complex field the corresponding rigidity problem is subtler. In particular, a detailed proof check shows that excluding line-supported laws alone is not enough: the relevant obstruction is support on a *generalized circle*, that is, on a line or on a circle. We isolate the correct one-coordinate rigidity statement, formulate a quantitative gap from the family of generalized circles, and deduce an equality-case theorem for countable sums of independent complex random variables. Combined with the usual compactness reduction, this yields Theorem 2.1: a dimension-free orthogonal lower bound under a uniform L^1–L^2 norm-equivalence hypothesis and a uniform quantitative separation from generalized-circle-supported laws.

1. Introduction

Stable phase retrieval asks whether vectors can be reconstructed, up to a global phase, from their magnitudes. In the setting of spans of independent random variables, the real theory reduces stability to a lower bound on

$$\||X_a| - |X_b|\|_{L^2}$$

for orthogonal unit coefficient vectors a and b. The compactness proof has a simple final step: if the lower bound fails, a limit argument produces two real random sums X and Y with $|X| = |Y|$ almost surely, hence $Y = \pm X$. Orthogonality then gives the contradiction.

Over the complex field the same compactness strategy remains natural, but the equality case becomes genuinely geometric. Equality of moduli only implies that Y is a pointwise phase multiple of X, and there is no reason for that phase to be constant a priori. One might try to recover the real proof by showing that support in the cone

$$\Gamma := \{(u, v) \in \mathbb{C}^2 : |u| = |v|\}$$

forces one marginal to lie on a real line. This is false: if Z_1 and Z_2 are independent Steinhaus variables, then

$$\operatorname{supp}(Z_1, Z_2) = \mathbb{T} \times \mathbb{T} \subset \Gamma,$$

yet neither marginal is line-supported.

A more careful audit shows a second, subtler point. The naive replacement "not supported on a line" is still too weak. Indeed, if

$$S = c + r\mathbb{T} \qquad (c \neq 0,\ r > 0),$$

then

$$|t - c| = r \qquad \text{for every } t \in S,$$

so support on a non-centered circle also creates exact modulus identities. Thus the correct geometric obstruction is support on a *generalized circle*, meaning either a real line or a circle in $\mathbb{C}$.

The first purpose of this paper is to isolate the right rigidity statement. We prove that if

$$|\alpha t + u| = |\beta t + v|$$

for every t in a set not contained in any generalized circle, then (β, v) must be a unimodular multiple of (α, u). Iterating this one-coordinate lemma yields an equality-case theorem for countable sums of independent complex random variables.

The second purpose is to record the correct quantitative nondegeneracy hypothesis. For line-supported laws there is a simple explicit quantity,

$$\mathrm{gap}_{\mathrm{line}}(\xi) := \inf_{\lambda\in\mathbb{T}} \|\mathrm{Im}(\lambda\xi)\|_{L^2}^2 = \frac{1-|\mathbb{E}[\xi^2]|}{2}$$

when $\mathbb{E}[|\xi|^2] = 1$. But the actual compactness argument needs a gap from the full family of generalized circles. We therefore define

$$\mathrm{gap}_{\mathrm{gc}}(\xi) := \inf_{\substack{A,C\in\mathbb{R},\ B\in\mathbb{C}\\ A^2+2|B|^2+C^2=1}} \|A|\xi|^2 + 2\,\mathrm{Re}(B\xi) + C\|_{L^1}.$$

This vanishes exactly when $\mathrm{supp}(\xi)$ is contained in a generalized circle.

Our main result is Theorem 2.1. It states that if independent coordinates satisfy a dimension-free L^1–L^2 norm-equivalence and a uniform lower bound $\mathrm{gap}_{\mathrm{gc}}(\xi_j) \geq \kappa$, then the orthogonal modulus gap is bounded below by a positive constant depending only on (A, B, κ), and in particular not on the dimension.

The paper is organized as follows. Section 2 states the main theorem. Section 3 contains the geometric rigidity statements. Section 4 proves Theorem 2.1 by combining the compactness reduction with the equality-case rigidity theorem. Appendix A records the compactness proposition in the exact form used in the proof.

2. Main result

Let (Ω, μ) be a probability space. For $n \in \mathbb{N}$, let $\xi_1, \ldots, \xi_n$ be independent complex-valued random variables on (Ω, μ). For $a = (a_1, \ldots, a_n) \in \mathbb{C}^n$, write

$$X_a := \sum_{j=1}^n a_j \xi_j.$$

Theorem 2.1 (Uniform complex orthogonal lower bound)**.** *Fix constants $0 < A \leq B < 1$ and $\kappa > 0$. Then there exists*

$$c = c(A, B, \kappa) > 0$$

such that the following holds for every $n \in \mathbb{N}$.

Let $\xi_1, \ldots, \xi_n$ be independent complex-valued random variables with

$$\mathbb{E}[\xi_j] = 0, \qquad \mathbb{E}[|\xi_j|^2] = 1 \qquad (1 \leq j \leq n),$$

and assume:

(1) *for every $u = (u_1, \ldots, u_n) \in \mathbb{C}^n$,*

$$A\|u\|_2 \leq \|\sum_{j=1}^n u_j \xi_j\|_{L^1(\mu)} \leq B\|u\|_2;$$

(2) *for every j,*

$$\mathrm{gap}_{\mathrm{gc}}(\xi_j) \geq \kappa.$$

Then for every $a, b \in \mathbb{C}^n$ with

$$\|a\|_2 = \|b\|_2 = 1, \qquad \langle a, b\rangle = 0,$$

one has

$$\||X_a| - |X_b|\|_{L^2(\mu)} \geq c.$$

Definition 2.2. For $f, g \in L^2(\mu)$, define the phase-invariant distance

$$d_{\mathbb{T}}(f, g) := \inf_{\lambda\in\mathbb{T}} \|f - \lambda g\|_{L^2(\mu)}.$$

Lemma 2.3 (Pointwise secant monotonicity)**.** *For $z, w \in \mathbb{C}$ and $t \in [0,1]$,*

$$\big||z+tw| - |z-tw|\big| \geq t\,\big||z+w| - |z-w|\big|.$$

Proof. Set

$$H(s) := |z+sw| + |z-sw| \qquad (s \in \mathbb{R}).$$

The function H is even and convex, since $s \mapsto |z+sw|$ is convex and $H(-s) = H(s)$. Therefore H is nondecreasing on $[0,\infty)$, so

$$H(t) \leq H(1).$$

On the other hand,

$$\big(|z+tw| - |z-tw|\big)\,H(t) = |z+tw|^2 - |z-tw|^2 = 4t\,\mathrm{Re}(z\overline{w}),$$

hence

$$\big||z+tw| - |z-tw|\big| = \frac{4t\,|\mathrm{Re}(z\overline{w})|}{H(t)} \geq \frac{4t\,|\mathrm{Re}(z\overline{w})|}{H(1)} = t\,\big||z+w| - |z-w|\big|. \qquad \square$$

Corollary 2.4 (Quantitative complex stable phase retrieval)**.** *Under the hypotheses of Theorem 2.1, there exists*

$$C = C(A,B,\kappa) > 0$$

such that for all

$$f, g \in E_n := \mathrm{span}\{\xi_1, \ldots, \xi_n\} \subset L^2(\mu),$$

one has

$$d_{\mathbb{T}}(f,g) \leq C\,\big\||f| - |g|\big\|_{L^2(\mu)}.$$

In particular, the space

$$\mathrm{span}\{\xi_1, \ldots, \xi_n\}$$

does stable phase retrieval over $\mathbb{C}$, with a stability constant depending only on (A,B,κ).

Proof. Let

$$U_n : \mathbb{C}^n \to E_n, \qquad U_n(a) := X_a.$$

Because the ξ_j are independent, centered, and unit-variance,

$$\langle U_n(a), U_n(b)\rangle_{L^2} = \mathbb{E}[X_a\overline{X_b}] = \sum_{j=1}^{n} a_j\overline{b_j} = \langle a, b\rangle.$$

Thus U_n is a unitary isomorphism.

Let $c = c(A,B,\kappa)$ be the constant from Theorem 2.1. We first show that if $a, b \in \mathbb{C}^n$ satisfy

$$\|a\|_2 = \|b\|_2 = r,$$

then

$$\inf_{\lambda\in\mathbb{T}} \|a - \lambda b\|_2 \leq \frac{\sqrt{2}}{c}\,\big\||X_a| - |X_b|\big\|_{L^2}.$$

If $r = 0$, this is trivial. Assume $r > 0$, and choose $\lambda \in \mathbb{T}$ so that

$$\inf_{\mu\in\mathbb{T}} \|a - \mu b\|_2 = \|a - \lambda b\|_2.$$

Replacing b by λb, we may assume that

$$\inf_{\mu\in\mathbb{T}} \|a - \mu b\|_2 = \|a - b\|_2 \qquad \text{and} \qquad \langle a, b\rangle \in [0,\infty).$$

Set

$$x := \frac{a}{r}, \qquad y := \frac{b}{r}, \qquad \alpha := \frac{\|x+y\|_2}{2}, \qquad \beta := \frac{\|x-y\|_2}{2}.$$

Then

$$x = \alpha u + \beta v, \qquad y = \alpha u - \beta v,$$

where

$$u := \frac{x+y}{2\alpha}, \qquad v := \frac{x-y}{2\beta}$$

when $\beta > 0$. If $\beta = 0$, then $x = y$ and there is nothing to prove. In the case $\beta > 0$, the choice $\langle x, y\rangle \in [0, \infty)$ gives

$$\langle u, v\rangle = \frac{\langle x+y, x-y\rangle}{4\alpha\beta} = 0,$$

and also

$$\alpha^2 - \beta^2 = \mathrm{Re}\langle x, y\rangle \geq 0,$$

so $0 \leq \beta/\alpha \leq 1$.

Let

$$U := X_u, \qquad V := X_v, \qquad t := \frac{\beta}{\alpha} \in [0, 1].$$

Pointwise,

$$\big||X_x| - |X_y|\big| = \alpha\,\big||U + tV| - |U - tV|\big|.$$

Lemma 2.3 yields

$$\big||X_x| - |X_y|\big| \geq \beta\,\big||U + V| - |U - V|\big|.$$

Now

$$p := \frac{u+v}{\sqrt{2}}, \qquad q := \frac{u-v}{\sqrt{2}}$$

are orthogonal unit vectors, so Theorem 2.1 gives

$$\big\||X_p| - |X_q|\big\|_{L^2} \geq c.$$

Since

$$X_p = \frac{U+V}{\sqrt{2}}, \qquad X_q = \frac{U-V}{\sqrt{2}},$$

we obtain

$$\big\||U+V| - |U-V|\big\|_{L^2} \geq \sqrt{2}\,c.$$

Therefore

$$\big\||X_x| - |X_y|\big\|_{L^2} \geq \sqrt{2}\,c\,\beta.$$

Recalling that

$$\|x - y\|_2 = 2\beta,$$

we conclude

$$\|x - y\|_2 \leq \frac{\sqrt{2}}{c}\,\big\||X_x| - |X_y|\big\|_{L^2}.$$

Multiplying by r gives the equal-norm estimate.

Now let $a, b \in \mathbb{C}^n$ be arbitrary, and set

$$G := \big\||X_a| - |X_b|\big\|_{L^2}.$$

Without loss of generality, $\|a\|_2 \geq \|b\|_2$. Write

$$r := \|a\|_2, \qquad s := \|b\|_2.$$

If $s = 0$, then

$$d_{\mathbb{T}}(X_a, X_b) = \|X_a\|_{L^2} = r = G,$$

so the desired estimate is immediate. Assume $s > 0$, and define

$$b' := \frac{r}{s}\,b.$$

Then

$$\|b'\|_2 = r, \qquad \big\||X_{b'}| - |X_b|\big\|_{L^2} = r - s.$$

By the reverse triangle inequality in L^2,

$$r - s = \big|\|X_a\|_{L^2} - \|X_b\|_{L^2}\big| \le \||X_a| - |X_b|\|_{L^2} = G.$$

Moreover,

$$d_{\mathbb{T}}(X_a, X_b) \le d_{\mathbb{T}}(X_a, X_{b'}) + \|X_{b'} - X_b\|_{L^2} \le d_{\mathbb{T}}(X_a, X_{b'}) + (r - s).$$

Applying the equal-norm estimate to a and b', and then using the triangle inequality for modulus differences, we get

$$d_{\mathbb{T}}(X_a, X_{b'}) \le \frac{\sqrt{2}}{c}\,\||X_a| - |X_{b'}|\|_{L^2} \le \frac{\sqrt{2}}{c}\,(G + r - s) \le \frac{2\sqrt{2}}{c}\,G.$$

Therefore

$$d_{\mathbb{T}}(X_a, X_b) \le \Big(\frac{2\sqrt{2}}{c} + 1\Big)G.$$

This proves the stated stability inequality with

$$C := \frac{2\sqrt{2}}{c(A, B, \kappa)} + 1.$$

□

Remark 2.5. The proof of Theorem 2.1 uses only the norm-equivalence bounds and the generalized-circle gap. The quantity $\mathrm{gap}_{\mathrm{line}}$ from Section 3 remains useful because it gives a completely explicit formula for the line-supported degeneracy, but the detailed proof check shows that line separation alone is not enough for the complex equality case.

3. Rigidity for the complex equality case

Remark 3.1. The implication

$$\mathrm{supp}(Z_1, Z_2) \subset \Gamma \quad \Longrightarrow \quad \text{one of } Z_1, Z_2 \text{ is line-supported}$$

is false. Indeed, if Z_1 and Z_2 are independent Steinhaus variables, then

$$\mathrm{supp}(Z_1, Z_2) = \mathbb{T} \times \mathbb{T} \subset \Gamma,$$

but neither marginal is supported on a real line $e^{i\theta}\mathbb{R}$.

Definition 3.2. A *generalized circle* in $\mathbb{C}$ means either a circle

$$\{z \in \mathbb{C} : |z - c| = r\}$$

with $c \in \mathbb{C}$ and $r \ge 0$, or a real line in $\mathbb{C} \simeq \mathbb{R}^2$.

Definition 3.3. For a complex-valued random variable $\xi \in L^2$, define its *line-gap* by

$$\mathrm{gap}_{\mathrm{line}}(\xi) := \inf_{\lambda \in \mathbb{T}} \|\mathrm{Im}(\lambda\xi)\|_{L^2}^2.$$

Definition 3.4. For a complex-valued random variable $\xi \in L^2$, define its *generalized-circle gap* by

$$\mathrm{gap}_{\mathrm{gc}}(\xi) := \inf_{\substack{A, C \in \mathbb{R},\ B \in \mathbb{C} \\ A^2 + 2|B|^2 + C^2 = 1}} \|A|\xi|^2 + 2\,\mathrm{Re}(B\xi) + C\|_{L^1}.$$

Remark 3.5. If ν is a probability measure on $\mathbb{C}$ with finite second moment, define

$$\mathrm{gap}_{\mathrm{gc}}(\nu) := \inf_{\substack{A, C \in \mathbb{R},\ B \in \mathbb{C} \\ A^2 + 2|B|^2 + C^2 = 1}} \int_{\mathbb{C}} |A|z|^2 + 2\,\mathrm{Re}(Bz) + C|\, d\nu(z).$$

If ξ has law ν, then $\mathrm{gap}_{\mathrm{gc}}(\xi) = \mathrm{gap}_{\mathrm{gc}}(\nu)$.

Lemma 3.6 (Classification of generalized circles)**.** *Let $A, C \in \mathbb{R}$ and $B \in \mathbb{C}$, not all zero. Then the zero set*

$$\{z \in \mathbb{C} : A|z|^2 + 2\,\mathrm{Re}(Bz) + C = 0\}$$

is a generalized circle.

Proof. If $A = 0$, then $B \neq 0$ or $C \neq 0$. If $B = 0$, the zero set is either empty or all of $\mathbb{C}$; the latter cannot occur because the coefficients are not all zero, so the zero set is empty. If $B \neq 0$, then the equation becomes

$$2\,\mathrm{Re}(Bz) + C = 0,$$

which is a real affine line in $\mathbb{C} \simeq \mathbb{R}^2$.

If $A \neq 0$, then completing the square gives

$$|z + \frac{B}{A}|^2 = \frac{|B|^2}{A^2} - \frac{C}{A}.$$

Hence the zero set is either empty or a circle. □

Lemma 3.7. *Let $\xi \in L^2$ satisfy $\mathbb{E}[|\xi|^2] = 1$. Then*

$$\mathrm{gap}_{\mathrm{line}}(\xi) = \frac{1 - |\mathbb{E}[\xi^2]|}{2}.$$

In particular, $\mathrm{gap}_{\mathrm{line}}(\xi) = 0$ if and only if ξ is supported on a real line $e^{i\theta}\mathbb{R}$.

Proof. For $\lambda \in \mathbb{T}$,

$$\|\mathrm{Im}(\lambda\xi)\|_{L^2}^2 = \frac{1}{2}\,\mathbb{E}[|\lambda\xi|^2] - \frac{1}{2}\,\mathrm{Re}(\mathbb{E}[(\lambda\xi)^2]) = \frac{1}{2} - \frac{1}{2}\,\mathrm{Re}(\lambda^2\mathbb{E}[\xi^2]).$$

Taking the infimum over $\lambda \in \mathbb{T}$ is the same as maximizing $\mathrm{Re}(\lambda^2\mathbb{E}[\xi^2])$, which yields

$$\mathrm{gap}_{\mathrm{line}}(\xi) = \frac{1 - |\mathbb{E}[\xi^2]|}{2}.$$

If $\mathrm{gap}_{\mathrm{line}}(\xi) = 0$, choose $\lambda \in \mathbb{T}$ minimizing the left-hand side. Then $\|\mathrm{Im}(\lambda\xi)\|_{L^2} = 0$, so $\mathrm{Im}(\lambda\xi) = 0$ almost surely and $\lambda\xi$ is real-valued. Thus ξ is supported on the real line $\overline{\lambda}\mathbb{R}$. The converse is immediate. □

Lemma 3.8. *For a complex-valued random variable $\xi \in L^2$,*

$$\mathrm{gap}_{\mathrm{gc}}(\xi) = 0$$

if and only if $\mathrm{supp}(\xi)$ *is contained in a generalized circle.*

Proof. If $\mathrm{supp}(\xi)$ is contained in a generalized circle, then there exist $A, C \in \mathbb{R}$ and $B \in \mathbb{C}$, not all zero, such that

$$A|z|^2 + 2\,\mathrm{Re}(Bz) + C = 0$$

for every $z \in \mathrm{supp}(\xi)$. After normalizing so that

$$A^2 + 2|B|^2 + C^2 = 1,$$

we obtain

$$A|\xi|^2 + 2\,\mathrm{Re}(B\xi) + C = 0 \qquad \text{a.s.},$$

hence $\mathrm{gap}_{\mathrm{gc}}(\xi) = 0$.

Conversely, if $\mathrm{gap}_{\mathrm{gc}}(\xi) = 0$, compactness of the coefficient sphere

$$\Big\{(A, B, C) : A, C \in \mathbb{R},\ B \in \mathbb{C},\ A^2 + 2|B|^2 + C^2 = 1\Big\}$$

and continuity of

$$(A, B, C) \longmapsto \|A|\xi|^2 + 2\,\mathrm{Re}(B\xi) + C\|_{L^1}$$

show that the infimum is attained. Thus there exist $A, C \in \mathbb{R}$ and $B \in \mathbb{C}$ with

$$A^2 + 2|B|^2 + C^2 = 1$$

such that

$$\|A|\xi|^2 + 2\,\mathrm{Re}(B\xi) + C\|_{L^1} = 0.$$

Hence

$$A|\xi|^2 + 2\,\mathrm{Re}(B\xi) + C = 0 \qquad \text{a.s.}$$

By Lemma 3.6, the zero set of this polynomial is a generalized circle, so $\mathrm{supp}(\xi)$ is contained in one. □

Remark 3.9. Excluding only centered circles is not enough. If

$$S = c + r\mathbb{T} \qquad (c \neq 0,\ r > 0),$$

then

$$|1 \cdot t - c| = |0 \cdot t + r| \qquad \text{for every } t \in S,$$

but the pairs $(1, -c)$ and $(0, r)$ are not unimodular multiples of one another.

Lemma 3.10 (One-coordinate generalized-circle rigidity)**.** *Let $S \subset \mathbb{C}$ be a set not contained in any generalized circle. Let $\alpha, \beta, u, v \in \mathbb{C}$, and assume that*

$$|\alpha t + u| = |\beta t + v| \qquad \textit{for every } t \in S.$$

Then

$$|\alpha| = |\beta|, \qquad \overline{\alpha}\, u = \overline{\beta}\, v.$$

Consequently, either $\alpha = \beta = 0$, or there exists $\lambda \in \mathbb{T}$ such that

$$\beta = \lambda\alpha, \qquad v = \lambda u.$$

Proof. Set

$$A := |\alpha|^2 - |\beta|^2 \in \mathbb{R}, \qquad B := \overline{\alpha}u - \overline{\beta}v \in \mathbb{C}, \qquad C := |u|^2 - |v|^2 \in \mathbb{R}.$$

Squaring the hypothesis gives

$$A|t|^2 + 2\,\mathrm{Re}(tB) + C = 0$$

for every $t \in S$.

If $A \neq 0$, then completing the square yields

$$|t + \frac{B}{A}|^2 = \frac{|B|^2}{A^2} - \frac{C}{A}$$

for every $t \in S$. Thus S is contained in a circle, contrary to hypothesis. Hence $A = 0$, which is

$$|\alpha| = |\beta|.$$

The identity therefore reduces to

$$2\,\mathrm{Re}(tB) + C = 0$$

for every $t \in S$. If $B \neq 0$, then S is contained in a real line, again contrary to hypothesis. Therefore $B = 0$, that is,

$$\overline{\alpha}\, u = \overline{\beta}\, v.$$

If $\alpha = 0$, then $\beta = 0$ by $|\alpha| = |\beta|$, and we are in the first alternative. Otherwise define $\lambda := \beta/\alpha \in \mathbb{T}$. Then

$$\overline{\alpha}u = \overline{\beta}v = \overline{\lambda}\,\overline{\alpha}v,$$

so $u = \overline{\lambda}v$, i.e.

$$v = \lambda u.$$

□

Lemma 3.11 (Support of an independent pair)**.** *Let U and V be independent random variables taking values in Polish spaces. Then*

$$\operatorname{supp}(U,V) = \operatorname{supp}(U) \times \operatorname{supp}(V).$$

Proof. If $u \in \operatorname{supp}(U)$ and $v \in \operatorname{supp}(V)$, then every neighborhood $\mathcal{U} \ni u$ and $\mathcal{V} \ni v$ satisfies

$$\mathbb{P}(U \in \mathcal{U},\ V \in \mathcal{V}) = \mathbb{P}(U \in \mathcal{U})\mathbb{P}(V \in \mathcal{V}) > 0,$$

so $(u,v) \in \operatorname{supp}(U,V)$. This proves

$$\operatorname{supp}(U) \times \operatorname{supp}(V) \subset \operatorname{supp}(U,V).$$

The reverse inclusion is immediate from the definition of support. □

Lemma 3.12 (Almost-sure vanishing extends to the support)**.** *Let S be a Polish space, let Z be an S-valued random variable, and let $F : S \to \mathbb{R}$ be continuous. If*

$$F(Z) = 0 \qquad \textit{almost surely,}$$

then

$$F(x) = 0 \qquad \textit{for every } x \in \operatorname{supp}(Z).$$

Proof. Assume that $x \in \operatorname{supp}(Z)$ and $F(x) \neq 0$. By continuity, there exist $\varepsilon > 0$ and an open neighborhood $\mathcal{U} \ni x$ such that

$$|F(y)| \geq \varepsilon \qquad (y \in \mathcal{U}).$$

Since $x \in \operatorname{supp}(Z)$, one has $\mathbb{P}(Z \in \mathcal{U}) > 0$. On that event,

$$|F(Z)| \geq \varepsilon,$$

contradicting the almost sure identity $F(Z) = 0$. □

Lemma 3.13 (Independent equality forces a constant)**.** *Let $U, V \in L^2$ be independent complex-valued random variables. If*

$$U = V \qquad \textit{almost surely,}$$

then there exists $c \in \mathbb{C}$ such that

$$U = c \qquad \textit{and} \qquad V = c$$

almost surely.

Proof. Since $U = V$ almost surely, the random variables U and V have the same law, hence

$$\mathbb{E}[U] = \mathbb{E}[V].$$

Independence gives

$$\mathbb{E}[U\overline{V}] = \mathbb{E}[U]\overline{\mathbb{E}[V]} = |\mathbb{E}[U]|^2.$$

On the other hand, $U = V$ almost surely implies

$$\mathbb{E}[U\overline{V}] = \mathbb{E}[|U|^2].$$

Therefore

$$\mathbb{E}[|U|^2] = |\mathbb{E}[U]|^2.$$

Equivalently,

$$\mathbb{E}[|U - \mathbb{E}[U]|^2] = 0.$$

Hence $U = \mathbb{E}[U]$ almost surely. Since $U = V$ almost surely, the same constant works for V. □

Lemma 3.14 (Vanishing independent sum)**.** *Let $(\zeta_j)_{j\geq 1}$ be independent centered complex-valued random variables with*

$$\mathbb{E}[|\zeta_j|^2] = 1 \qquad (j \geq 1).$$

Let $c = (c_j)_{j\geq 1} \in \ell^2$, and define

$$Z := \sum_{j\geq 1} c_j \zeta_j$$

with convergence in L^2. If

$$Z = 0 \qquad \textit{almost surely,}$$

then

$$c_j = 0 \qquad (j \geq 1).$$

Proof. For $N \in \mathbb{N}$, set

$$Z^{(N)} := \sum_{j=1}^{N} c_j \zeta_j.$$

Then $Z^{(N)} \to Z$ in L^2. By independence, centering, and unit variance,

$$\mathbb{E}[|Z^{(N)}|^2] = \sum_{j=1}^{N} |c_j|^2.$$

Passing to the limit and using $Z = 0$ almost surely gives

$$0 = \mathbb{E}[|Z|^2] = \sum_{j\geq 1} |c_j|^2.$$

Hence every coefficient c_j vanishes. □

Corollary 3.15 (Rigidity for countable independent sums)**.** *Let $(\zeta_j)_{j\geq 1}$ be independent complex-valued random variables, and let $a = (a_j)_{j\geq 1}, b = (b_j)_{j\geq 1} \in \ell^2$. Define*

$$X := \sum_{j\geq 1} a_j \zeta_j, \qquad Y := \sum_{j\geq 1} b_j \zeta_j,$$

with convergence in L^2. Assume $|X| = |Y|$ almost surely. If, for every j, the support of ζ_j is not contained in any generalized circle, then there exists $\lambda \in \mathbb{T}$ such that

$$Y = \lambda X \qquad \textit{almost surely.}$$

Proof. If $a = 0$ and $b = 0$, then $X = Y = 0$ almost surely, so the conclusion is trivial. Hence we may assume that $(a, b) \neq (0, 0)$.

Choose k such that $(a_k, b_k) \neq (0, 0)$. Write

$$X = a_k \zeta_k + U, \qquad Y = b_k \zeta_k + V,$$

where

$$U := \sum_{j\neq k} a_j \zeta_j, \qquad V := \sum_{j\neq k} b_j \zeta_j.$$

Because ζ_k is independent of (U, V), Lemma 3.11 gives

$$\operatorname{supp}(\zeta_k, U, V) = \operatorname{supp}(\zeta_k) \times \operatorname{supp}(U, V).$$

The continuous function

$$F(t, u, v) := |a_k t + u|^2 - |b_k t + v|^2$$

vanishes almost surely at (ζ_k, U, V). By Lemma 3.12, it therefore vanishes on the whole support of (ζ_k, U, V).

Fix $(u, v) \in \operatorname{supp}(U, V)$. Then

$$|a_k t + u| = |b_k t + v| \qquad \text{for every } t \in \operatorname{supp}(\zeta_k).$$

By Lemma 3.10,

$$|a_k| = |b_k|, \qquad \overline{a_k}u = \overline{b_k}v.$$

Since $(a_k, b_k) \neq (0,0)$, the equality $|a_k| = |b_k|$ shows that both coefficients are nonzero. Set

$$\lambda := \frac{b_k}{a_k} \in \mathbb{T}.$$

Then

$$v = \lambda u \qquad \text{for every } (u,v) \in \operatorname{supp}(U,V).$$

Thus $V = \lambda U$ almost surely, and therefore

$$Y = b_k\zeta_k + V = \lambda a_k\zeta_k + \lambda U = \lambda X$$

almost surely. □

Corollary 3.16 (Affine rigidity relative to an independent base)**.** *Let $(\zeta_j)_{j\geq 1}$ be independent centered complex-valued random variables satisfying*

$$\mathbb{E}[|\zeta_j|^2] = 1 \qquad (j \geq 1),$$

let $a = (a_j)_{j\geq 1}, b = (b_j)_{j\geq 1} \in \ell^2$, and let W, Z be complex-valued random variables independent of the family $(\zeta_j)_{j\geq 1}$. Define

$$X := W + \sum_{j\geq 1} a_j\zeta_j, \qquad Y := Z + \sum_{j\geq 1} b_j\zeta_j,$$

with convergence in L^2. Assume $|X| = |Y|$ almost surely. If, for every j, the support of ζ_j is not contained in any generalized circle, and if $(a,b) \neq (0,0)$, then there exists $\lambda \in \mathbb{T}$ such that

$$b_j = \lambda a_j \qquad (j \geq 1), \qquad \text{and} \qquad Z = \lambda W \quad \text{almost surely.}$$

In particular,

$$Y = \lambda X \qquad \text{almost surely.}$$

Proof. Choose k such that $(a_k, b_k) \neq (0,0)$. Write

$$X = a_k\zeta_k + U, \qquad Y = b_k\zeta_k + V,$$

where

$$U := W + \sum_{j\neq k} a_j\zeta_j, \qquad V := Z + \sum_{j\neq k} b_j\zeta_j.$$

Because ζ_k is independent of (U,V), Lemma 3.11 gives

$$\operatorname{supp}(\zeta_k, U, V) = \operatorname{supp}(\zeta_k) \times \operatorname{supp}(U,V).$$

The continuous function

$$F(t,u,v) := |a_k t + u|^2 - |b_k t + v|^2$$

vanishes almost surely at (ζ_k, U, V). By Lemma 3.12, it therefore vanishes on the whole support of (ζ_k, U, V).

Fix $(u,v) \in \operatorname{supp}(U,V)$. Then

$$|a_k t + u| = |b_k t + v| \qquad \text{for every } t \in \operatorname{supp}(\zeta_k).$$

By Lemma 3.10,

$$|a_k| = |b_k|, \qquad \overline{a_k}u = \overline{b_k}v.$$

Since $(a_k, b_k) \neq (0,0)$, both coefficients are nonzero. Set

$$\lambda := \frac{b_k}{a_k} \in \mathbb{T}.$$

Then $v = \lambda u$ for every $(u,v) \in \operatorname{supp}(U,V)$, hence

$$V = \lambda U \qquad \text{almost surely.}$$

Therefore

$$Z + \sum_{j\neq k} b_j\zeta_j = \lambda W + \sum_{j\neq k} \lambda a_j\zeta_j \qquad \text{almost surely.}$$

Rearranging,

$$Z - \lambda W = \sum_{j\neq k}(\lambda a_j - b_j)\zeta_j \qquad \text{almost surely.}$$

The left-hand side and the right-hand side are independent, because the former is measurable with respect to $\sigma(W, Z)$ and the latter is measurable with respect to $\sigma(\zeta_j : j \neq k)$, and these sigma-algebras are independent. Since they are also almost surely equal, Lemma 3.13 shows that both sides are almost surely equal to the same constant $c \in \mathbb{C}$. Taking expectations and using $\mathbb{E}[\zeta_j] = 0$ gives $c = 0$. Thus

$$Z = \lambda W \qquad \text{almost surely,}$$

and

$$\sum_{j\neq k}(\lambda a_j - b_j)\zeta_j = 0 \qquad \text{almost surely.}$$

Lemma 3.14 therefore yields

$$\lambda a_j - b_j = 0 \qquad (j \neq k).$$

Hence $b_j = \lambda a_j$ for all $j \neq k$. The same identity holds for $j = k$ by definition of λ, so $Y = \lambda X$ almost surely. □

4. Proof of the main theorem

Proof of Theorem 2.1. Assume, for contradiction, that the conclusion is false. Then there exist integers n_m, independent families

$$\xi_1^{(m)}, \dots, \xi_{n_m}^{(m)},$$

and orthogonal unit vectors $a^{(m)}, b^{(m)} \in \mathbb{C}^{n_m}$ such that

$$\||X_{a^{(m)}}| - |X_{b^{(m)}}|\|_{L^2} \to 0,$$

while every family satisfies the hypotheses of the theorem with the same constants (A, B, κ).

Apply the compactness reduction recorded in Appendix A. After passing to a subsequence, Proposition A.1 provides independent centered unit-variance complex random variables $(\zeta_j)_{j\geq 1}$, coefficient sequences $a, b \in \ell^2$ with

$$\|a\|_2 = \|b\|_2 = 1, \qquad \langle a, b\rangle = 0,$$

and an exact modulus identity

$$|X| = |Y| \qquad \text{almost surely,}$$

where

$$X := \sum_{j\geq 1} a_j\zeta_j, \qquad Y := \sum_{j\geq 1} b_j\zeta_j.$$

Moreover, for every finitely supported $u \in \ell^2$,

$$A\|u\|_2 \leq \|\sum_{j\geq 1} u_j\zeta_j\|_{L^1} \leq B\|u\|_2,$$

and

$$\mathrm{gap}_{\mathrm{gc}}(\zeta_j) \geq \kappa \qquad (j \geq 1).$$

Fix j. By Lemma 3.8, the lower bound $\mathrm{gap}_{\mathrm{gc}}(\zeta_j) \geq \kappa > 0$ implies that $\mathrm{supp}(\zeta_j)$ is not contained in any generalized circle.

Corollary 3.15 therefore yields a phase $\lambda \in \mathbb{T}$ such that

$$Y = \lambda X \qquad \text{almost surely.}$$

Using independence, centering, and unit variance, we have

$$\mathbb{E}[X\overline{Y}] = \sum_{j\geq 1} a_j \overline{b_j} = \langle a, b\rangle = 0.$$

On the other hand,

$$\mathbb{E}[X\overline{Y}] = \overline{\lambda}\,\mathbb{E}[|X|^2] = \overline{\lambda}\sum_{j\geq 1}|a_j|^2 = \overline{\lambda}\|a\|_2^2 = \overline{\lambda},$$

which is impossible since $\lambda \in \mathbb{T}$.

This contradiction proves Theorem 2.1. □

Appendix A. The compactness reduction used in Theorem 2.1

The proof of Theorem 2.1 uses exactly one compactness input. We record it here in the precise form needed in Section 4.

Proposition A.1 (Compactness reduction)**.** *Fix $0 < A \leq B < 1$ and $\kappa > 0$. Let*

$$(\xi_1^{(m)}, \dots, \xi_{n_m}^{(m)}), \qquad a^{(m)}, b^{(m)} \in \mathbb{C}^{n_m}, \qquad m \geq 1,$$

be a sequence such that, for each m,

$$\mathbb{E}[\xi_j^{(m)}] = 0, \qquad \mathbb{E}[|\xi_j^{(m)}|^2] = 1 \qquad (1 \leq j \leq n_m),$$

$$A\|u\|_2 \leq \|\sum_{j=1}^{n_m} u_j \xi_j^{(m)}\|_{L^1} \leq B\|u\|_2 \qquad (u \in \mathbb{C}^{n_m}),$$

$$\mathrm{gap}_{\mathrm{gc}}(\xi_j^{(m)}) \geq \kappa \qquad (1 \leq j \leq n_m),$$

and

$$\|a^{(m)}\|_2 = \|b^{(m)}\|_2 = 1, \qquad \langle a^{(m)}, b^{(m)}\rangle = 0.$$

Assume finally that

$$\||X_{a^{(m)}}| - |X_{b^{(m)}}|\|_{L^2} \to 0.$$

Then, after passing to a subsequence, there exist:

(1) *independent centered unit-variance complex random variables $(\zeta_j)_{j\geq 1}$;*
(2) *coefficient sequences $a, b \in \ell^2$ with*

$$\|a\|_2 = \|b\|_2 = 1, \qquad \langle a, b\rangle = 0;$$

(3) *the exact modulus identity*

$$|\sum_{j\geq 1} a_j\zeta_j| = |\sum_{j\geq 1} b_j\zeta_j| \qquad \textit{almost surely;}$$

(4) *the same norm-equivalence bounds*

$$A\|u\|_2 \leq \|\sum_{j\geq 1} u_j\zeta_j\|_{L^1} \leq B\|u\|_2$$

for every finitely supported $u \in \ell^2$;
(5) *the same generalized-circle gap lower bound*

$$\mathrm{gap}_{\mathrm{gc}}(\zeta_j) \geq \kappa \qquad (j \geq 1).$$

Remark A.2 (Formalization interface). From the point of view of a Lean formalization, Proposition A.1 is the only compactness input used in the proof of Theorem 2.1. Once this proposition is available, the proof of Theorem 2.1 uses only the algebraic rigidity lemmas from Section 3 and the finitely stated reductions in the rest of Appendix A.

Lemma A.3 (Ordered coefficient bound)**.** *After simultaneously reordering the coordinates of each model so that*

$$\max(|a_1^{(m)}|, |b_1^{(m)}|) \geq \max(|a_2^{(m)}|, |b_2^{(m)}|) \geq \cdots ,$$

one has

$$\max(|a_j^{(m)}|, |b_j^{(m)}|) \leq \sqrt{\frac{2}{j}} \qquad (j, m \geq 1).$$

Proof. Let

$$c_j^{(m)} := \max(|a_j^{(m)}|, |b_j^{(m)}|).$$

Since the $c_j^{(m)}$ are nonincreasing in j,

$$j\,(c_j^{(m)})^2 \leq \sum_{i=1}^{j} (c_i^{(m)})^2 \leq \sum_{i=1}^{n_m} (|a_i^{(m)}|^2 + |b_i^{(m)}|^2) = 2.$$

□

Lemma A.4 (Diagonal extraction of one-coordinate profiles)**.** *After passing to a subsequence, there exist complex numbers a_j, b_j and probability measures ν_j on $\mathbb{C}$ such that for every fixed j,*

$$a_j^{(m)} \to a_j, \qquad b_j^{(m)} \to b_j, \qquad \mathrm{Law}(\xi_j^{(m)}) \Rightarrow \nu_j$$

as $m \to \infty$. Moreover,

$$\sum_{j=1}^{\infty} |a_j|^2 \leq 1, \qquad \sum_{j=1}^{\infty} |b_j|^2 \leq 1.$$

Proof. For each fixed j, the coefficient sequences

$$(a_j^{(m)})_{m\geq 1}, \qquad (b_j^{(m)})_{m\geq 1}$$

are bounded by 1. The one-coordinate laws

$$\nu_j^{(m)} := \mathrm{Law}(\xi_j^{(m)})$$

form a tight family because

$$\int_{\mathbb{C}} |z|^2 \, d\nu_j^{(m)}(z) = 1.$$

By a diagonal subsequence argument, we may therefore assume that for every fixed j,

$$a_j^{(m)} \to a_j, \qquad b_j^{(m)} \to b_j, \qquad \nu_j^{(m)} \Rightarrow \nu_j.$$

For each N,

$$\sum_{j=1}^{N} |a_j|^2 = \lim_{m\to\infty} \sum_{j=1}^{N} |a_j^{(m)}|^2 \leq 1.$$

Taking the supremum over N gives $\sum_j |a_j|^2 \leq 1$. The proof for b is the same. □

Lemma A.5 (Linear-growth integral continuity)**.** *Let μ_m, μ be probability measures on $\mathbb{C}$ such that*

$$\mu_m \Rightarrow \mu \qquad \textit{and} \qquad \sup_m \int_{\mathbb{C}} |z|^2 \, d\mu_m(z) < \infty.$$

Then for every continuous function $f : \mathbb{C} \to \mathbb{C}$ satisfying

$$|f(z)| \leq C_f(1 + |z|) \qquad (z \in \mathbb{C})$$

for some constant C_f, one has

$$\int_{\mathbb{C}} f(z) \, d\mu_m(z) \to \int_{\mathbb{C}} f(z) \, d\mu(z).$$

Proof. Fix $\varepsilon > 0$, and set

$$M := \sup_m \int_{\mathbb{C}} |z|^2 \, d\mu_m(z) < \infty.$$

If $C_f = 0$, then $f = 0$ identically and there is nothing to prove. Choose $R \geq 1$ so large that

$$\frac{2M}{R} < \frac{\varepsilon}{2C_f}.$$

By Portmanteau,

$$\int_{\mathbb{C}} |z|^2 \, d\mu(z) \leq \liminf_{m\to\infty} \int_{\mathbb{C}} |z|^2 \, d\mu_m(z) \leq M.$$

Hence, for every m,

$$\mu_m(\{|z| > R\}) \leq \frac{M}{R^2}, \qquad \int_{\{|z|>R\}} |z| \, d\mu_m(z) \leq \sqrt{M}\, \mu_m(\{|z| > R\})^{1/2} \leq \frac{M}{R},$$

and the same estimates hold for μ. Therefore

$$\int_{\{|z|>R\}} (1 + |z|) \, d\mu_m(z) < \frac{\varepsilon}{2C_f}, \qquad \int_{\{|z|>R\}} (1 + |z|) \, d\mu(z) < \frac{\varepsilon}{2C_f}.$$

Choose a continuous cutoff $\eta_R : \mathbb{C} \to [0, 1]$ such that

$$\eta_R(z) = 1 \text{ for } |z| \leq R, \qquad \eta_R(z) = 0 \text{ for } |z| \geq 2R.$$

Then $f\eta_R$ is bounded and continuous, so weak convergence, applied to the real and imaginary parts, gives

$$\int f\eta_R \, d\mu_m \to \int f\eta_R \, d\mu.$$

Moreover,

$$\Big|\int f(1 - \eta_R) \, d\mu_m\Big| \leq C_f \int_{\{|z|>R\}} (1 + |z|) \, d\mu_m(z) < \frac{\varepsilon}{2},$$

and the same bound holds for μ. Hence

$$\limsup_{m\to\infty} \Big|\int f \, d\mu_m - \int f \, d\mu\Big| \leq \varepsilon.$$

Since $\varepsilon > 0$ was arbitrary, the proof is complete. □

Lemma A.6 (Finite-head convergence)**.** *Let*

$$(\Omega_\infty, \mu_\infty) := \prod_{j\geq 1} (\mathbb{C}, \nu_j),$$

and let ζ_j be the coordinate random variables. Then the ζ_j are independent and centered, with

$$\mathbb{E}[|\zeta_j|^2] \leq 1 \qquad (j \geq 1).$$

For every $N \in \mathbb{N}$, define

$$X_m^{(N)} := \sum_{j=1}^N a_j^{(m)} \xi_j^{(m)}, \qquad Y_m^{(N)} := \sum_{j=1}^N b_j^{(m)} \xi_j^{(m)},$$

and

$$X^{(N)} := \sum_{j=1}^N a_j \zeta_j, \qquad Y^{(N)} := \sum_{j=1}^N b_j \zeta_j.$$

Then

$$(X_m^{(N)}, Y_m^{(N)}) \Rightarrow (X^{(N)}, Y^{(N)})$$

in distribution on $\mathbb{C}^2$*. Moreover, for every* $u \in \mathbb{C}^N$*,*

$$A\|u\|_2 \le \|\sum_{j=1}^N u_j\zeta_j\|_{L^1(\mu_\infty)} \le B\|u\|_2.$$

Proof. For each j, Lemma A.5 applied to

$$f(z) := z$$

shows that

$$\int_{\mathbb{C}} z\, d\nu_j(z) = \lim_{m\to\infty}\int_{\mathbb{C}} z\, d\nu_j^{(m)}(z) = 0.$$

Thus ζ_j is centered. Since $z \mapsto |z|^2$ is nonnegative and lower semicontinuous, Portmanteau gives

$$\mathbb{E}[|\zeta_j|^2] = \int_{\mathbb{C}} |z|^2\, d\nu_j(z) \le \liminf_{m\to\infty}\int_{\mathbb{C}} |z|^2\, d\nu_j^{(m)}(z) = 1.$$

Weak convergence of the one-coordinate laws implies weak convergence of the finite product laws

$$\nu_1^{(m)} \otimes \cdots \otimes \nu_N^{(m)} \Rightarrow \nu_1 \otimes \cdots \otimes \nu_N.$$

Since the coefficients also converge coordinatewise, the continuous mapping theorem gives

$$(X_m^{(N)}, Y_m^{(N)}) \Rightarrow (X^{(N)}, Y^{(N)}).$$

Now fix $u \in \mathbb{C}^N$. Independence and unit variance imply

$$\mathbb{E}\Big[\Big|\sum_{j=1}^N u_j\xi_j^{(m)}\Big|^2\Big] = \sum_{j=1}^N |u_j|^2,$$

so the laws of

$$\sum_{j=1}^N u_j\xi_j^{(m)}$$

have uniformly bounded second moments. Since these random variables converge in distribution to

$$\sum_{j=1}^N u_j\zeta_j,$$

Lemma A.5, applied to $f(z) = |z|$, gives

$$\|\sum_{j=1}^N u_j\xi_j^{(m)}\|_{L^1} \to \|\sum_{j=1}^N u_j\zeta_j\|_{L^1}.$$

Passing to the limit in

$$A\|u\|_2 \le \|\sum_{j=1}^N u_j\xi_j^{(m)}\|_{L^1} \le B\|u\|_2$$

gives the stated estimate. □

Lemma A.7 (Quadratic-affine integral continuity)**.** *Let* μ_m, μ *be probability measures on* $\mathbb{C}$ *such that*

$$\mu_m \Rightarrow \mu \qquad \text{and} \qquad \int_{\mathbb{C}} |z|^2\, d\mu_m(z) \to \int_{\mathbb{C}} |z|^2\, d\mu(z) < \infty.$$

Then for every $A, C \in \mathbb{R}$ *and* $B \in \mathbb{C}$*,*

$$\int_{\mathbb{C}} |A|z|^2 + 2\operatorname{Re}(Bz) + C|\, d\mu_m(z) \longrightarrow \int_{\mathbb{C}} |A|z|^2 + 2\operatorname{Re}(Bz) + C|\, d\mu(z).$$

Proof. Set

$$f(z) := |A|z|^2 + 2\,\mathrm{Re}(Bz) + C|.$$

There is a constant $C_0 = C_0(A, B, C)$ such that

$$f(z) \le C_0(1 + |z|^2) \qquad (z \in \mathbb{C}).$$

If $C_0 = 0$, then $f = 0$ identically and the claim is trivial. Fix $\varepsilon > 0$. Since μ has finite second moment, we may choose $R \ge 1$ so large that $\mu(\{|z| = R\}) = 0$ and

$$\int_{\{|z|>R\}} (1 + |z|^2)\, d\mu(z) < \frac{\varepsilon}{4C_0}.$$

Let

$$\phi_R(z) := \min(|z|^2, R^2).$$

Then ϕ_R is bounded and continuous, so weak convergence gives

$$\int_{\mathbb{C}} \phi_R\, d\mu_m \to \int_{\mathbb{C}} \phi_R\, d\mu.$$

Since $\mu(\{|z| = R\}) = 0$, Portmanteau gives

$$\mu_m(\{|z| > R\}) \to \mu(\{|z| > R\}).$$

For any probability measure σ on $\mathbb{C}$, set

$$\tau_R(\sigma) := \int_{\mathbb{C}} |z|^2\, d\sigma - \int_{\mathbb{C}} \phi_R\, d\sigma + R^2\sigma(\{|z| > R\}).$$

Then

$$\tau_R(\sigma) = \int_{\{|z|>R\}} |z|^2\, d\sigma(z),$$

so the convergence of second moments, together with the two convergences above, implies

$$\int_{\{|z|>R\}} |z|^2\, d\mu_m(z) = \tau_R(\mu_m) \to \tau_R(\mu) = \int_{\{|z|>R\}} |z|^2\, d\mu(z).$$

Hence, for all sufficiently large m,

$$\mu_m(\{|z| > R\}) < \frac{\varepsilon}{2C_0} \qquad \text{and} \qquad \int_{\{|z|>R\}} |z|^2\, d\mu_m(z) < \frac{\varepsilon}{2C_0}.$$

$$\mu(\{|z| > R\}) < \frac{\varepsilon}{4C_0}$$

already holds because $\mu(\{|z| > R\}) \le \int_{\{|z|>R\}} (1 + |z|^2)\, d\mu$. Now choose a continuous cutoff $\eta_R : \mathbb{C} \to [0, 1]$ such that

$$\eta_R(z) = 1 \text{ for } |z| \le R, \qquad \eta_R(z) = 0 \text{ for } |z| \ge 2R.$$

Then $f\eta_R$ is bounded and continuous, so

$$\int f\eta_R\, d\mu_m \to \int f\eta_R\, d\mu.$$

Moreover, for all sufficiently large m,

$$\int f(1 - \eta_R)\, d\mu_m \le C_0\mu_m(\{|z| > R\}) + C_0 \int_{\{|z|>R\}} |z|^2\, d\mu_m < \varepsilon,$$

and the same bound holds for μ. Hence

$$\limsup_{m\to\infty} \Big| \int f\, d\mu_m - \int f\, d\mu \Big| \le 2\varepsilon.$$

Since $\varepsilon > 0$ was arbitrary, the proof is complete. □

Lemma A.8 (Passage of the generalized-circle gap)**.** *Fix j. Assume in addition that*

$$\int_{\mathbb{C}} |z|^2 \, d\nu_j(z) = 1.$$

Then the limit law ν_j from Lemma A.4 satisfies

$$\mathrm{gap}_{\mathrm{gc}}(\nu_j) \geq \kappa.$$

Proof. For every m,

$$\int_{\mathbb{C}} |z|^2 \, d\nu_j^{(m)}(z) = 1,$$

and by hypothesis the same holds for ν_j. Therefore Lemma A.7 applies to the sequence $\nu_j^{(m)} \Rightarrow \nu_j$. If $\mathrm{gap}_{\mathrm{gc}}(\nu_j) < \kappa$, choose a normalized triple (A, B, C) such that

$$\int_{\mathbb{C}} |A|z|^2 + 2\,\mathrm{Re}(Bz) + C| \, d\nu_j(z) < \kappa.$$

For all sufficiently large m, the same strict inequality holds for $\nu_j^{(m)}$, contradicting the hypothesis $\mathrm{gap}_{\mathrm{gc}}(\xi_j^{(m)}) \geq \kappa$. □

Theorem A.9 (No-loss of mass)**.** *After the same reordering and subsequence extraction as above, one has*

$$\lim_{N\to\infty} \sup_m \sum_{j>N} (|a_j^{(m)}|^2 + |b_j^{(m)}|^2) = 0.$$

Moreover, the same compactness theorem gives

$$\int_{\mathbb{C}} |z|^2 \, d\nu_j(z) = 1 \qquad (j \geq 1).$$

Remark A.10 (Lean-oriented decomposition of Theorem A.9). The proof sketch below can be formalized by isolating four explicit targets.

(1) *Second-profile extraction.* If tail vanishing fails, then after passing to a subsequence and reindexing a tail block, one extracts a nonzero profile $(\alpha_j), (\beta_j) \in \ell^2$ together with independent variables $(\widetilde{\zeta}_j)$ and an exact identity

$$|X^{[1]} + \widetilde{X}| = |Y^{[1]} + \widetilde{Y}| \qquad \text{a.s.}$$

(2) *Inheritance of coordinate hypotheses.* Every extracted profile retains centeredness, unit variance, and the lower bound $\mathrm{gap}_{\mathrm{gc}} \geq \kappa$ on each coordinate law.

(3) *Phase coherence of profiles.* Corollary 3.16 applied recursively to each new profile forces a single phase $\lambda \in \mathbb{T}$ such that

$$\beta^{[r]} = \lambda \alpha^{[r]}$$

for every nonzero profile r.

(4) *Mass contradiction.* The profile decomposition records both

$$\sum_r \|\alpha^{[r]}\|_2^2 = 1 \qquad \text{and} \qquad \sum_r \langle \alpha^{[r]}, \beta^{[r]} \rangle = 0.$$

After inserting $\beta^{[r]} = \lambda \alpha^{[r]}$, this becomes

$$0 = \overline{\lambda} \sum_r \|\alpha^{[r]}\|_2^2,$$

which is impossible.

For a Lean development, these four items are a more precise replacement for the single phrase "profile-decomposition theorem".

Proof sketch. This is the only step that is not fully reproved here. It is the compact profile-decomposition theorem from the companion real paper, and it is exactly the compact analogue of the head/tail dichotomy that is made explicit in the quantitative real proof.

If the displayed tail vanishing failed, then a positive amount of coefficient mass would remain in arbitrarily far tails. By Lemma A.3, those tail coefficients are individually small. The compactness proof then organizes that persistent tail mass into a second profile. More precisely, one chooses integers $N_m \to \infty$ so that the truncated tails beyond N_m carry a fixed positive proportion of the ℓ^2-mass, reindexes those tail coordinates so that they start at 1, and repeats the diagonal extraction on that reindexed tail model. This produces a new nonzero profile

$$(\alpha_j)_{j\geq 1}, \qquad (\beta_j)_{j\geq 1},$$

with corresponding independent variables $(\widetilde{\zeta}_j)$, and an exact limit identity of the form

$$|X^{[1]} + \widetilde{X}| = |Y^{[1]} + \widetilde{Y}| \qquad \text{a.s.},$$

where $X^{[1]}, Y^{[1]}$ are the first extracted profile sums and $\widetilde{X}, \widetilde{Y}$ come from the escaping tail profile. The construction uses disjoint coordinate blocks, so the sigma-algebra generated by $(X^{[1]}, Y^{[1]})$ is independent of that generated by $(\widetilde{\zeta}_j)_{j\geq 1}$. Moreover the reindexed tail variables are still centered and unit-variance and retain the same generalized-circle gap lower bound κ, because those properties are coordinatewise and are preserved under subsequences and reindexing.

At that point the proof splits into the same two alternatives as in the real argument:

(1) either one signed tail is asymptotically concentrated, which forces a rigid head relation;
(2) or both signed tails are diffuse, in which case symmetrization and anticoncentration give a positive lower bound for the modulus defect.

Either alternative contradicts the assumption

$$\||X_{a^{(m)}}| - |X_{b^{(m)}}|\|_{L^2} \to 0.$$

The role of Section 3 is to kill the possibility of multiple profiles. Applying Corollary 3.16 to the second profile, with the first profile playing the role of the independent base (W, Z), gives a unimodular λ such that

$$\widetilde{Y} = \lambda \widetilde{X} \qquad \text{and} \qquad Y^{[1]} = \lambda X^{[1]}.$$

Equivalently,

$$\beta_j = \lambda \alpha_j \qquad (j \geq 1)$$

for the second profile coefficients, and the same phase relation holds between the first profile coefficients. If one extracts a third escaping profile, the same argument applied to the identity obtained after removing the first two profiles shows that it must carry the same phase λ. Iterating, every nonzero profile satisfies

$$\beta^{[r]} = \lambda \alpha^{[r]}.$$

The profile theorem also records that the profiles come from disjoint coordinate blocks and that their squared norms add up to the total coefficient mass. In particular,

$$\sum_r \|\alpha^{[r]}\|_2^2 = 1, \qquad \sum_r \|\beta^{[r]}\|_2^2 = 1, \qquad \sum_r \langle \alpha^{[r]}, \beta^{[r]} \rangle = 0.$$

Summing the orthogonality relations over all nonzero profiles then gives

$$0 = \sum_r \langle \alpha^{[r]}, \beta^{[r]} \rangle = \overline{\lambda} \sum_r \|\alpha^{[r]}\|_2^2,$$

which is impossible because the total ℓ^2-mass is 1. Hence only one profile can survive, which is exactly the statement that no mass is lost at infinity.

The same profile theorem also keeps track of the one-coordinate second moments and shows that no second-moment mass is lost in the fixed-coordinate profiles:

$$\int_{\mathbb{C}} |z|^2 \, d\nu_j(z) = 1 \qquad (j \geq 1).$$

□

Lemma A.11 (Nonnegative zero expectation)**.** *Let $Z \in L^1$ satisfy*

$$Z \geq 0 \qquad \textit{almost surely.}$$

If

$$\mathbb{E}[Z] = 0,$$

then

$$Z = 0 \qquad \textit{almost surely.}$$

Proof. If $\mathbb{P}(Z > 0) > 0$, then for some $n \in \mathbb{N}$,

$$\mathbb{P}\Big(Z \geq \frac{1}{n}\Big) > 0.$$

Hence

$$\mathbb{E}[Z] \geq \frac{1}{n}\,\mathbb{P}\Big(Z \geq \frac{1}{n}\Big) > 0,$$

a contradiction. □

Lemma A.12 (Vanishing modulus defect passes to the limit)**.** *Let $(X_m, Y_m)_{m\geq 1}$ and (X, Y) be $\mathbb{C}^2$-valued random variables. Assume that*

$$(X_m, Y_m) \Rightarrow (X, Y), \qquad \mathbb{E}\big[\big||X_m| - |Y_m|\big|\big] \to 0,$$

and

$$\sup_m \mathbb{E}\big[\big(|X_m| - |Y_m|\big)^2\big] < \infty.$$

Then

$$|X| = |Y| \qquad \textit{almost surely.}$$

Proof. Set

$$Z_m := \big||X_m| - |Y_m|\big|, \qquad Z := \big||X| - |Y|\big|.$$

By the continuous mapping theorem,

$$Z_m \Rightarrow Z.$$

The L^2-bound gives

$$\sup_m \mathbb{E}[|Z_m|^2] < \infty.$$

Lemma A.5, applied to $f(z) = z$, therefore gives

$$\mathbb{E}[Z] = 0.$$

Lemma A.11 now yields $Z = 0$ almost surely, which is exactly the claim. □

Lemma A.13 (Tail vanishing implies the limit model)**.** *Assume the conclusion of Theorem A.9. Then the conclusion of Proposition A.1 follows.*

Proof. By Lemma A.4, the limit coefficient sequences $a = (a_j)$ and $b = (b_j)$ lie in ℓ^2. Let ζ_j be the coordinate variables on $(\Omega_\infty, \mu_\infty)$.

We first prove that

$$\|a\|_2 = \|b\|_2 = 1, \qquad \langle a, b\rangle = 0.$$

Fix $\varepsilon > 0$. By Theorem A.9, choose N so large that

$$\sup_m \sum_{j>N} |a_j^{(m)}|^2 < \varepsilon \qquad \text{and} \qquad \sup_m \sum_{j>N} |b_j^{(m)}|^2 < \varepsilon.$$

Then for every m,

$$\sum_{j=1}^{N} |a_j^{(m)}|^2 > 1 - \varepsilon.$$

Passing to the limit in m,

$$\sum_{j=1}^{N} |a_j|^2 \geq 1 - \varepsilon.$$

Since $\sum_j |a_j|^2 \leq 1$, this gives $\|a\|_2 = 1$. The same argument gives $\|b\|_2 = 1$.

For orthogonality, write

$$0 = \sum_{j=1}^{n_m} a_j^{(m)} \overline{b_j^{(m)}} = \sum_{j=1}^{N} a_j^{(m)} \overline{b_j^{(m)}} + \sum_{j>N} a_j^{(m)} \overline{b_j^{(m)}}.$$

By Cauchy–Schwarz,

$$\Big| \sum_{j>N} a_j^{(m)} \overline{b_j^{(m)}} \Big| \leq \Big(\sum_{j>N} |a_j^{(m)}|^2 \Big)^{1/2} \Big(\sum_{j>N} |b_j^{(m)}|^2 \Big)^{1/2} < \varepsilon.$$

Therefore

$$\Big| \sum_{j=1}^{N} a_j^{(m)} \overline{b_j^{(m)}} \Big| < \varepsilon.$$

Passing to the limit in m and then letting $N \to \infty$ gives $\langle a, b \rangle = 0$.

Define

$$X := \sum_{j \geq 1} a_j \zeta_j, \qquad Y := \sum_{j \geq 1} b_j \zeta_j,$$

with convergence in $L^2(\mu_\infty)$. This follows from independence, the bound $\mathbb{E}[|\zeta_j|^2] \leq 1$, and $a, b \in \ell^2$.

We next show that

$$(X_{a^{(m)}}, X_{b^{(m)}}) \Rightarrow (X, Y)$$

in distribution on $\mathbb{C}^2$. Fix a bounded Lipschitz function $\Phi : \mathbb{C}^2 \to \mathbb{R}$, and write

$$X_m := X_{a^{(m)}}, \qquad Y_m := X_{b^{(m)}}.$$

For $N \in \mathbb{N}$, let $X_m^{(N)}, Y_m^{(N)}, X^{(N)}, Y^{(N)}$ be as in Lemma A.6. Then

$$\big| \mathbb{E}[\Phi(X_m, Y_m)] - \mathbb{E}[\Phi(X, Y)] \big|$$

is bounded by the sum of

$$\big| \mathbb{E}[\Phi(X_m, Y_m)] - \mathbb{E}[\Phi(X_m^{(N)}, Y_m^{(N)})] \big|,$$
$$\big| \mathbb{E}[\Phi(X_m^{(N)}, Y_m^{(N)})] - \mathbb{E}[\Phi(X^{(N)}, Y^{(N)})] \big|,$$

and

$$\big| \mathbb{E}[\Phi(X^{(N)}, Y^{(N)})] - \mathbb{E}[\Phi(X, Y)] \big|.$$

The middle term tends to 0 as $m \to \infty$ by Lemma A.6. For the first term, the Lipschitz bound and the upper L^1–L^2 estimate yield

$$\mathbb{E}[|X_m - X_m^{(N)}|] \leq B \Big(\sum_{j>N} |a_j^{(m)}|^2 \Big)^{1/2},$$

and similarly for $Y_m - Y_m^{(N)}$. By Theorem A.9, these bounds go to 0 uniformly in m as $N \to \infty$. For the limit variables, independence, centering, and the identity $\mathbb{E}[|\zeta_j|^2] = 1$ give

$$\mathbb{E}[|X - X^{(N)}|^2] = \sum_{j>N} |a_j|^2, \qquad \mathbb{E}[|Y - Y^{(N)}|^2] = \sum_{j>N} |b_j|^2.$$

Therefore

$$\mathbb{E}[|X - X^{(N)}|] \le \Big(\sum_{j>N} |a_j|^2\Big)^{1/2}, \qquad \mathbb{E}[|Y - Y^{(N)}|] \le \Big(\sum_{j>N} |b_j|^2\Big)^{1/2},$$

and these quantities tend to 0 as $N \to \infty$ because $a, b \in \ell^2$. Hence $(X_m, Y_m) \Rightarrow (X, Y)$.

Since

$$\||X_m| - |Y_m|\|_{L^2} \to 0,$$

we have

$$\mathbb{E}\big[\big||X_m| - |Y_m|\big|\big] \to 0.$$

Lemma A.12 therefore gives

$$|X| = |Y| \qquad \text{almost surely.}$$

Finally, Theorem A.9 gives

$$\mathbb{E}[|\zeta_j|^2] = 1 \qquad (j \ge 1).$$

Applying Lemma A.8 to the law ν_j therefore yields

$$\mathrm{gap}_{\mathrm{gc}}(\zeta_j) = \mathrm{gap}_{\mathrm{gc}}(\nu_j) \ge \kappa \qquad (j \ge 1).$$

Lemma A.6 already shows that the ζ_j are centered. Thus the ζ_j are centered and unit-variance, and Lemma A.6 gives the finite-support L^1–L^2 bounds. This is precisely the conclusion of Proposition A.1. □

Proof of Proposition A.1. Simultaneously reorder coordinates in each model as in Lemma A.3. Then apply Lemmas A.4–A.8, Theorem A.9, and Lemma A.13. □